RAJALAKSHMI ENGINEERING COLLEGE

Jaya Raghul A 251801110-2025-AIDS

Started on	Thursday, 24 September 2026, 10:25 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:52 AM
Time taken	26 mins 56 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    int slow = a[0];
    int fast = a[0];

    do {
        slow = a[slow];
        fast = a[a[fast]];
    } while (slow != fast);
    slow = a[0];
    while (slow != fast) {
        slow = a[slow];
        fast = a[fast];
    }
    printf("%d\n", slow);
    return 0;
}

```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Jaya Raghul A 251801110-2025-AIDS

J2

Started on	Thursday, 24 September 2026, 10:52 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:25 AM
Time taken	32 mins 33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    int a[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    int slow = a[0];
    int fast = a[0];
    do {
        slow = a[slow];
        fast = a[a[fast]];
    } while (slow != fast);
    slow = a[0];
    while (slow != fast) {
        slow = a[slow];
        fast = a[fast];
    }
    printf("%d\n", slow );

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Done

Closes: Sunday, 11 October 2026, 8:25 AM

Attempts allowed: 10

Grading method: Highest grade

Summary of your previous attempts

Attempt	State	Marks / 1.00	Grade / 10.00	Review
1	Finished Submitted Thursday, 8 October 2026, 10:34 AM	1.00	10.00	Review

Highest grade: 10.00 / 10.00.

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RAJALAKSHMI ENGINEERING COLLEGE

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J2

Started on	Thursday, 8 October 2026, 11:04 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:05 AM
Time taken	1 min 38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int N1, N2;

        scanf("%d", &N1);

        int a[N1];
        for (int i = 0; i < N1; i++) {
            scanf("%d", &a[i]);
        }

        scanf("%d", &N2);

        int b[N2];
        for (int i = 0; i < N2; i++) {
            scanf("%d", &b[i]);
        }
```

```
        }

        int i = 0, j = 0;

        while (i < N1 && j < N2) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j]) {
                i++;
            }
            else {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Jaya Raghul A 251801110-2025-AIDS

J2

Started on	Thursday, 8 October 2026, 11:06 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:06 AM
Time taken	33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, k;

    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    scanf("%d", &k);

    int i = 0, j = 1;

    while (i < n && j < n) {

        if (i == j) {
            j++;
        }
        else if (a[j] - a[i] == k) {
            printf("1\n");
            return 0;
        }
        else if (a[j] - a[i] < k) {
            j++;
        }
        else {
            i++;
        }
    }

    printf("0\n");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Jaya Raghul A 251801110-2025-AIDS

J2

Started on	Thursday, 8 October 2026, 11:06 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:07 AM
Time taken	39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main()
{
    int n, k;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    scanf("%d", &k);

    int i = 0, j = 1;

    while (i < n && j < n)
    {
        if (i == j)
        {
            j++;
        }
        else if (a[j] - a[i] == k)
        {
            printf("1");
            return 0;
        }
        else if (a[j] - a[i] < k)
        {
            j++;
        }
        else
        {
            i++;
        }
    }

    printf("0");
```

```
40      printf( 0 );
41
42      return 0;
43  }
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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